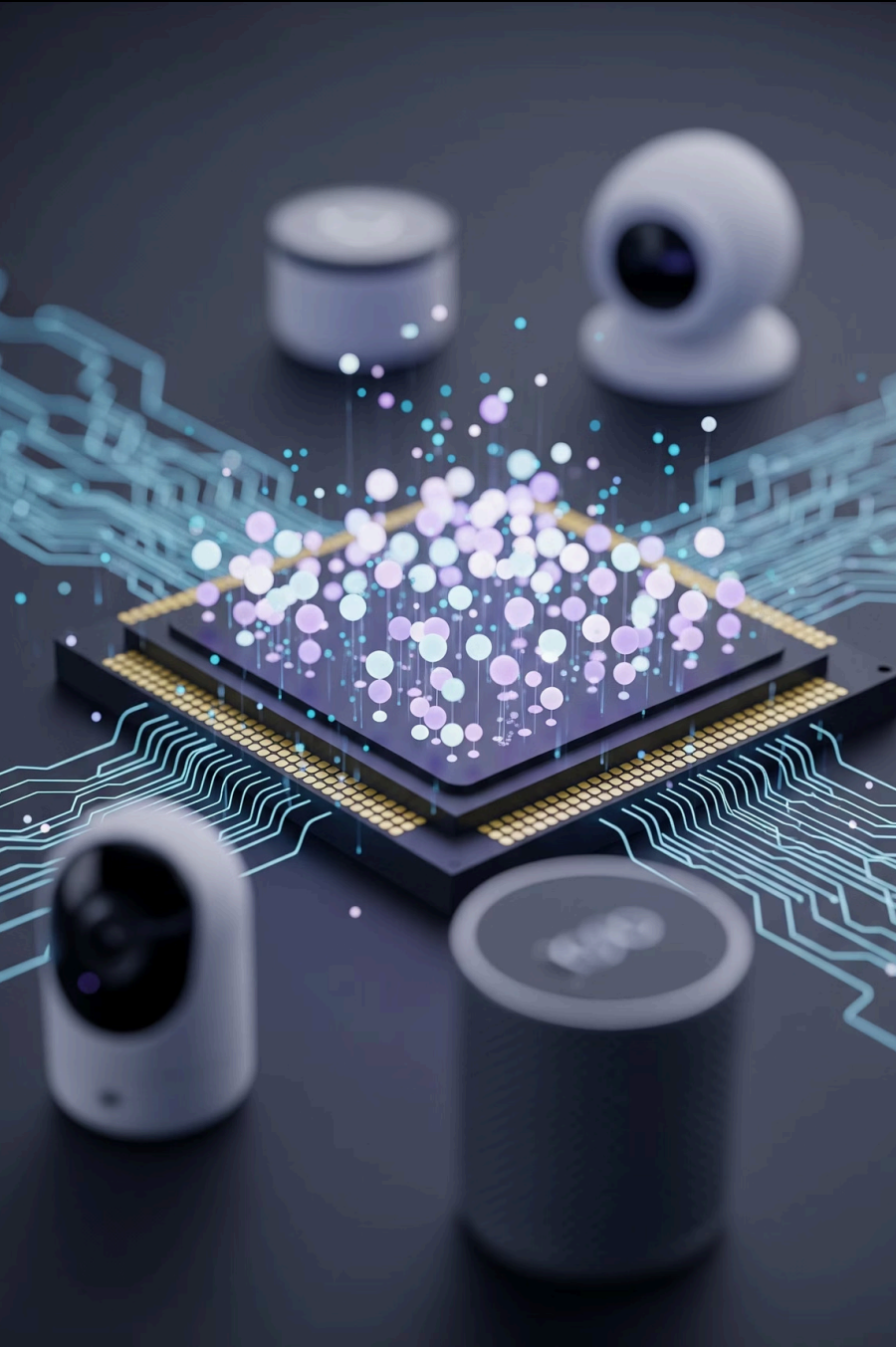




What Is Edge AI and Why It Matters

Bringing Intelligence Closer to Data & Users

A strategic approach for organizations seeking faster, more private, and cost-effective AI solutions

Edge AI Defined

Running ML on/near the data source

Edge AI deploys machine learning algorithms directly on devices, gateways, or on-premises servers instead of relying solely on cloud infrastructure

Real-world examples

Cameras, robots, kiosks, vehicles, factory lines, retail point-of-sale systems, and hospital room monitoring equipment

Key advantage

Dramatically reduced round trips to the cloud, enabling faster response times and more efficient processing



The Four Drivers of Edge AI

<100ms

Latency

Critical for safety systems
and seamless user
experiences requiring near-
instantaneous reactions

TB/day

Bandwidth

Avoid streaming raw video
and sensor data that creates
network bottlenecks

GDPR

Privacy

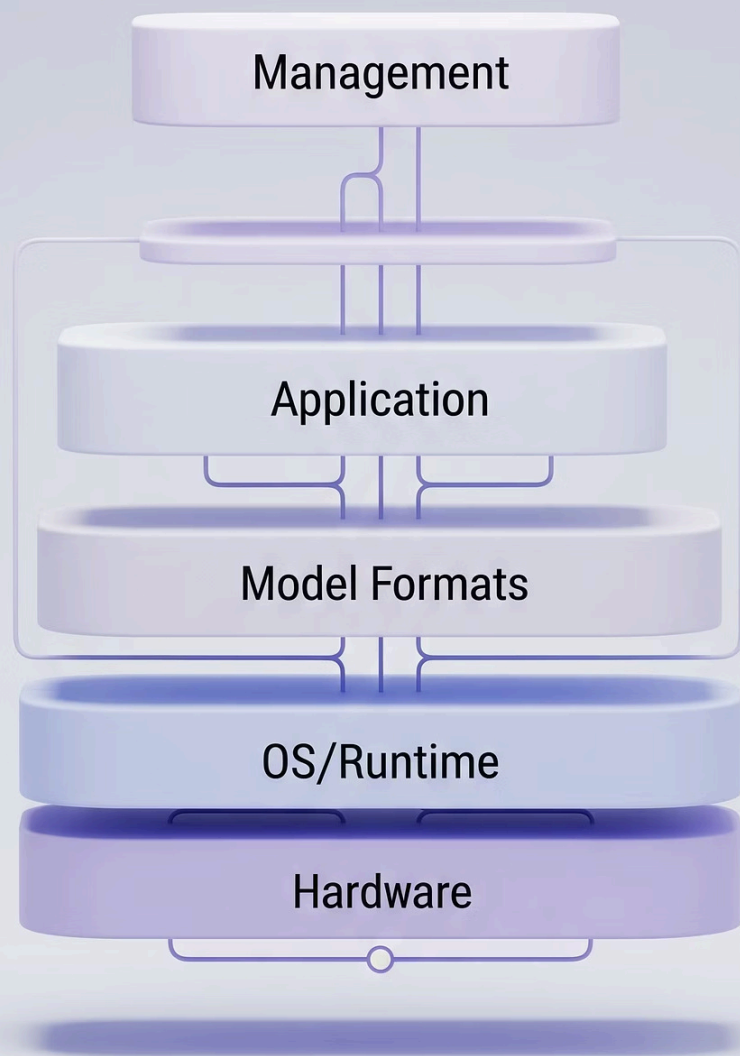
Keep sensitive data local to
meet compliance
requirements and enhance
security

99.9%

Resilience

Maintain operations during
outages while reducing
cloud egress costs

These drivers create compelling business cases for Edge AI deployment across industries



Typical Edge AI Technology Stack

1

Hardware

- NVIDIA Jetson
- x86 + GPU/TPU/NPU
- ARM SoCs
- Industrial PCs

2

OS/Runtime

- Linux
- JetPack
- CUDA, OpenVINO
- Core ML, NNAPI

3

Model Format

- ONNX
- TensorRT engine
- TFLite
- Core ML

4

App & Management

- gRPC/REST services
- Local message bus
- OTA updates
- Fleet orchestration

Where Edge AI Shines: Use Cases



Vision

Defect detection, checkout-free retail, safety PPE compliance monitoring



Audio/NLP

Wake-word detection, multilingual offline speech recognition, call-center kiosks



Industrial/IoT

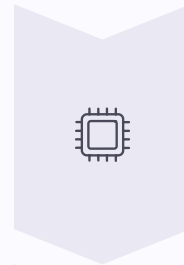
Predictive maintenance, anomaly detection in real-time



Mobility

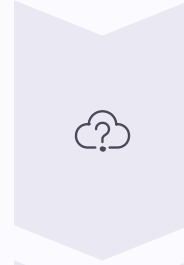
Autonomous navigation, vehicle-to-everything (V2X), advanced driver assistance

Edge vs. Cloud: Complementary Roles



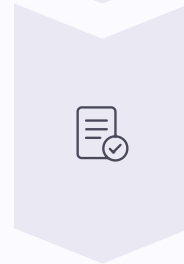
Edge

Handles real-time inference, filtering, and pre-aggregation of data closer to the source



Cloud

Manages heavy training workloads, global coordination, model registry, and complex analytics



Synchronization

Edge sends telemetry up; cloud sends optimized models and policies down

The optimal solution leverages the strengths of both environments



Navigating Edge Constraints



Compute/Memory/Power Limitations

Edge devices often operate with strict resource constraints, requiring model optimization



Intermittent Connectivity

Systems must be designed with offline-first principles to maintain operations during network outages



Heterogeneous Hardware

Diverse device ecosystem requires portable solutions and specialized compilers



Physical Security Concerns

Devices deployed in the field need protection against tampering and data theft

Successful Edge AI implementations require strategies to address these unique challenges

Model Optimization Toolkit



Optimization techniques can reduce model size by up to 90% while maintaining acceptable accuracy levels

Quantization

Convert floating-point operations to INT8/FP16 using TensorRT, TFLite, or OpenVINO

Pruning & Distillation

Create smaller, faster models that maintain similar accuracy through selective parameter reduction

Accelerator-aware Compilers

Leverage TensorRT, TVM, and XLA to optimize for specific hardware targets

Efficient Pipelines

Implement operation fusion, zero-copy memory transfers, and asynchronous I/O

Deploying & Managing Edge AI at Scale

Packaging

Containerize models and application code using Docker or create lightweight bundles optimized for resource-constrained devices

Orchestration

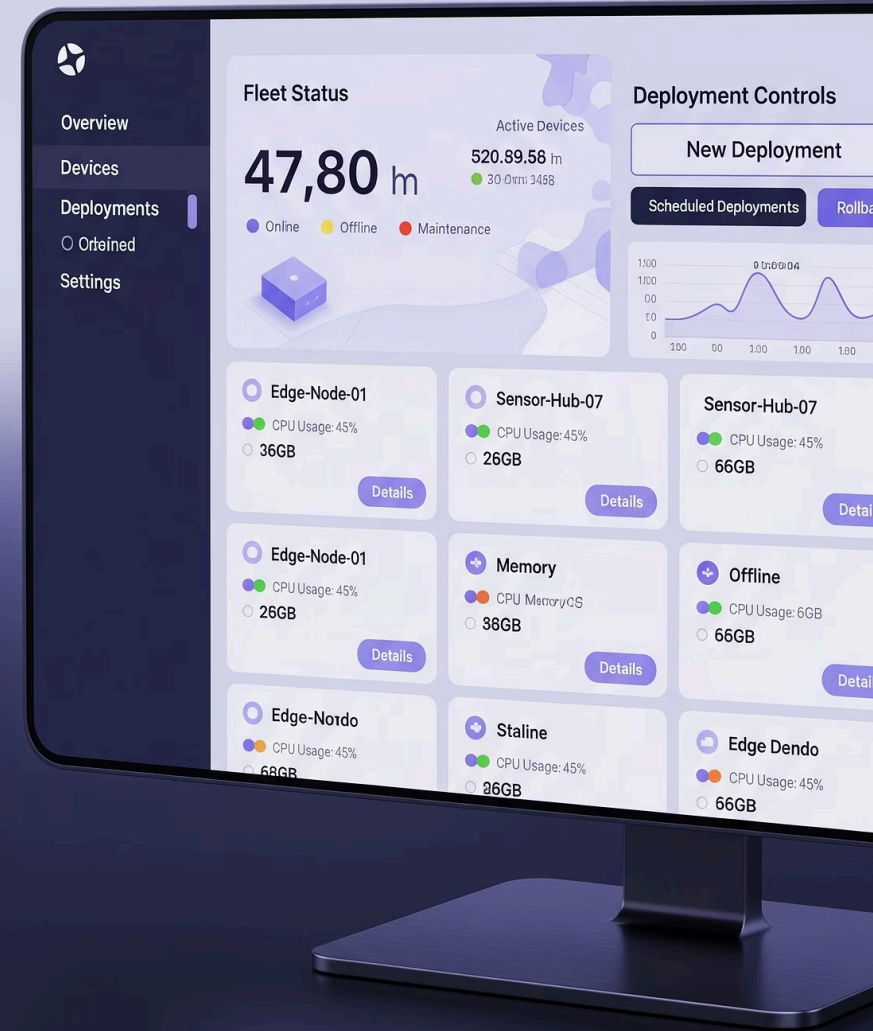
Deploy using lightweight Kubernetes variants (K3s/MicroK8s) or specialized fleet managers like Balena or Azure IoT Edge

Updates

Implement blue/green deployment strategies with staged rollouts and automatic rollback capabilities

Observability

Collect local logs and metrics, then forward through gateways to cloud platforms using Prometheus or OpenTelemetry



Edge AI Security Essentials



Hardware Security



Implement secure boot with TPM/TEE (Trusted Platform Module/Trusted Execution Environment) and full disk encryption

Communication Security



Use mutual TLS for device-to-cloud connections with regular certificate rotation

Identity Management



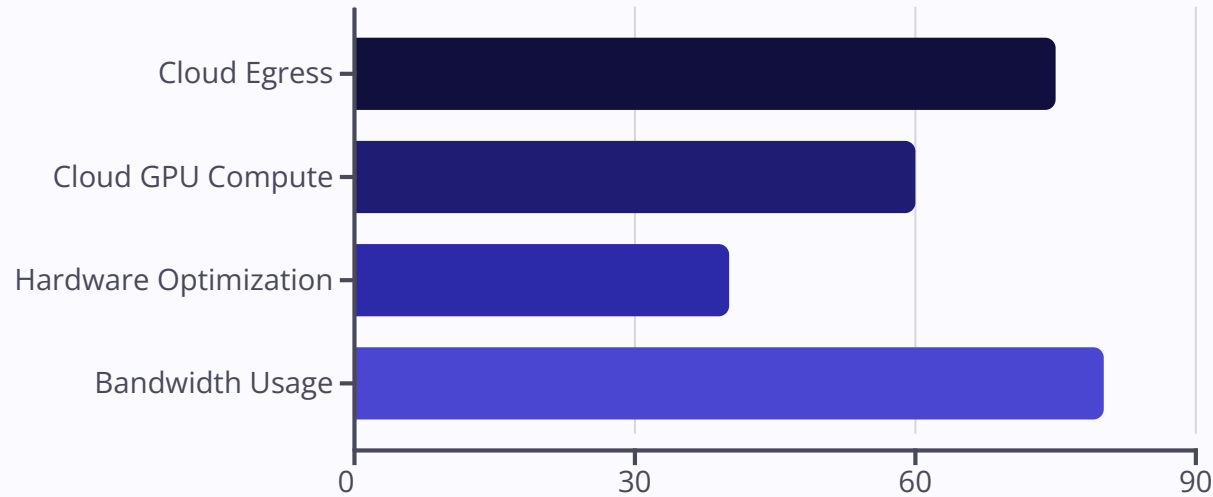
Apply least-privilege principles with scoped tokens for device identities

Supply Chain Security



Verify signed models and containers with Software Bill of Materials (SBOM) and provenance tracking

Cost & ROI Considerations



Cost Reduction Strategies

Save on egress and cloud GPU costs by performing inference locally

Hybrid Analytics

Upload processed features and events rather than raw media

Hardware Efficiency

Reduce hardware costs by right-sizing accelerators for specific workloads

Key Metrics

Measure ROI through latency SLOs, accuracy, uptime, and cost per inference event



Key Takeaways

Edge AI delivers **real-time, private, resilient**, and cost-effective intelligence



Design for Constraints

Optimize models for limited resources and ensure reliable offline operation



Embrace Hybrid Approach

Leverage edge and cloud together for training, model management, and fleet operations



Prioritize Infrastructure

Invest in security, observability, and update strategies from day one

Start your Edge AI journey with a small proof-of-concept that addresses a specific business challenge